

Yingru (Richard) Li

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Reinforcement Learning · LLM · Optimization · Reasoning & Agents

EXPERIENCE

xAI - SpaceX

Member of Technical Staff

Singapore

December 2025 – Present

- **Make core contributions to large-scale (“big-run”) reinforcement learning for the Grok models.**
 - *Science* — stability, efficiency, and the core algorithms, recipes, optimizer and scaling laws; helped identify and resolve critical training instabilities that benefit every stage of the RL pipeline, contributing to one of the longest stable RL runs to date.
 - *Long Horizon* — contributed to training recipes for long-horizon agentic RL.
 - *On-policy distillation* — contributed to training recipes and research infrastructure.
 - *Infrastructure* — built and helped scale these training recipes in the large-scale distributed RL systems.

ByteDance

LLM Algorithm Research Scientist

Singapore

2025 – December 2025

- **Led research on RL training stability and efficiency for LLM reasoning and agents.**
 - Diagnosed and formalized the *RL Training Collapse* phenomenon, tracing instability to implicit off-policy-ness caused by the training–inference mismatch; developed Trust Region Masking (ICML’26), Optimal Token Baseline (ICML’26), Dynamic Vocabulary Pruning, and a simple LR-scheduling fix.
 - Core contributor to **veRL** (a leading open-source RL framework, ~18k GitHub stars): authored the Roll-out Correction framework for general off-policy training and contributed the Optimal Token Baseline estimator (recipes and training infrastructure).
 - Co-led project on end-to-end RL training for multi-turn tool-integrated reasoning (SimpleTIR, ICLR’26) and Triton kernel code generation agent (Dr. Kernel, ICML’26).

Tencent AI & Robotics X

Student Research Staff Member, Agent Center

Shenzhen, China

2019 – 2022

- Led research for *scalable exploration in deep RL*: HyperAgent (ICML’24) and HyperDQN (ICLR’22), and *stable off-policy learning*: Divergence-Augmented Policy Optimization (NeurIPS’19).

Microsoft Research

Research Intern, Theory Center

Beijing, China

2016

- Research on combinatorial optimization and reinforcement / bandit learning algorithms.

SELECTED PUBLICATIONS

Full list at [Google Scholar](#). ★ denotes equal contribution; † denotes corresponding author.

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**
Yingru Li, Jiawei Xu, Ziniu Li, Jiakai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026*.
2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**
Yingru Li, Jiakai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026*.
3. **Optimistic Thompson Sampling for No-Regret Learning in Unknown Games**
Yingru Li, Liangqi Liu, Wenqiang Pu, Hao Liang, Zhi-Quan Luo. *IEEE Transactions on Signal Processing*, 2026.
4. **On the Scaling Flaws of Verifier-Guided Beam Search in Mathematical Reasoning**
Fei Yu, Yingru Li, Benyou Wang. *Transactions on Machine Learning Research (TMLR)*, 2026.
5. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**
Wei Liu, Jiawei Xu, Yingru Li, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026*.
6. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**

- Jiawei Wang, Jiakai Liu, Yuqian Fu, **Yingru Li**, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026*.
7. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026*.
 8. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
Jiakai Liu*, **Yingru Li***[†], Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Blog*, 2025.
 9. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**
Yaxiang Zhang*, **Yingru Li***[†], Jiakai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *arXiv 2026*.
 10. **Scalable Exploration via Ensemble++**
Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025*.
 11. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**
Yingru Li, Jiawei Xu, Jiakai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025*.
 12. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
Meng kang Hu, Yuhang Zhou, Wendong Fan, ..., **Yingru Li**, ..., Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025*.
 13. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024*.
 14. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
Yingru Li, Zhi-Quan Luo. *AISTATS 2024*.
 15. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
Ziniu Li, **Yingru Li**[†], Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022*.
 16. **Divergence-Augmented Policy Optimization**
Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS 2019*.

EDUCATION

The Chinese University of Hong Kong (CUHK) 2025

Ph.D. in Computer and Information Engineering.

Advisor: *Zhi-Quan (Tom) Luo*. Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST) 2020

B.E./M.E. in Computer Science. Ranked 1st/134 overall; 1st/26 in Software and Theoretical CS.

SELECTED HONORS

- National Scholarship (China) — highest national academic honor, top 0.2% nationwide, 2018.
- Best Student Paper, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- SRIBD Ph.D. Fellowship (Gold Class), 2023; Presidential Ph.D. Fellowship, CUHK, 2019–2023.
- Qiming Star Award — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.
- First Prize, China National Mathematics Olympiad (Province-level Math League), 2012.